Innovative Science Olympiad

Model Paper 2025 – 2026

Grade 8

Total Questions: 50 | **Time: 1.5 hr.** | **Total Marks: 100**

Syllabus

Cell - Structure and Functions, Microorganisms, Crop Production and Management, Conservation of Plants and Animals, Reproduction in Animals, Coal and Petroleum, Combustion and Flame, Materials – Metals and Non-metals, Force and Pressure, Friction, Sound, Chemical Effects of Electric Current, Light.

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Science

1. A mechanic uses a wrench (spanner) of length 20 cm to loosen a nut. He applies a force of 50 N at the end of the handle. Calculate the moment of force generated.

(A) 1000 Nm (B) 10 Nm

(C) 100 Nm (D) 2.5 Nm

2. Imagine you are an astronaut taking a sealed bag of potato chips from sea level to the top of a high mountain. What will happen to the bag and why?

(A) The bag will shrink because atmospheric pressure is higher at the top.

(B) The bag will remain unchanged as plastic is rigid.

(C) The bag will puff up (expand) because atmospheric pressure decreases at higher altitudes.

(D) The bag will expand because the temperature is hotter at the top.

3. A food technologist is studying the preservation of fruit juices. She observes that if juice is left in an open container, it starts smelling like alcohol and bubbles of gas are produced. Upon microscopic examination, she finds a unicellular fungus. Identify the process occurring and the gas being released.

(A) Aerobic respiration; Oxygen gas

(B) Pasteurization; Water vapor

(C) Fermentation; Carbon dioxide gas

(D) Nitrogen fixation; Nitrogen gas

4. Consider two organisms: 'X' is a frog and 'Y' is a human. Which of the following statements correctly compares the fertilization and development processes in these two organisms?

(A) Both X and Y undergo internal fertilization, but only Y shows vivipary.

(B) X undergoes external fertilization and is oviparous, while Y undergoes internal fertilization and is viviparous.

(C) X undergoes internal fertilization and is oviparous, while Y undergoes external fertilization and is viviparous.

(D) Both X and Y undergo external fertilization, but X shows indirect development.

5. **Assertion:** Hydrogen is considered a very efficient fuel but is not commonly used as a domestic fuel.
Reason: Hydrogen has a very high calorific value but is difficult to handle and transport safely.
(A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(C) Assertion is true but Reason is false.

(D) Assertion is false but Reason is true.

6. A student performs an experiment where she takes two airtight containers. In Container A, she places moist bread. In Container B, she places dry, toasted bread. After five days, she observes a fuzzy, thread-like growth in Container A but nothing in Container B. What does this experiment prove about the growth requirements of fungi like Rhizopus?
(A) Fungi require carbon dioxide from the air to grow.

(B) Fungi can only grow in dark environments with no airflow.

(C) Moisture is a critical requirement for the germination of fungal spores.

(D) Toasting bread creates a chemical that acts as a permanent antifungal agent.

7. In the design of high-speed vehicles like bullet trains and racing cars, the bodies are "streamlined". Which specific type of friction is being minimized here, and what is the mechanism?
(A) Rolling friction; by reducing the contact area of tires.

(B) Fluid friction (Drag); by reducing the resistance offered by air molecules.

(C) Static friction; by making the surface extremely smooth.

(D) Sliding friction; by using magnetic levitation.

8. A student is studying the "Budding" process in Hydra. She makes the following observations. Which of these observations is scientifically accurate regarding asexual reproduction?

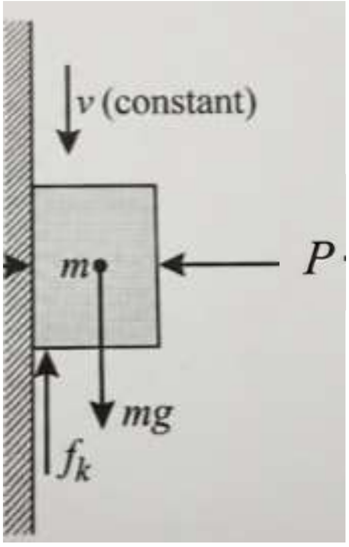
(A) The offspring is genetically different from the parent due to the fusion of two buds.

(B) A small bulge appears on the parent body, develops into a tiny individual, and then detaches.

(C) Hydra produces eggs that are fertilized by the buds of another Hydra.

(D) The parent Hydra splits into two equal halves, and each half becomes a new individual.

9. Imagine a block of mass m sliding down a rough vertical wall at a constant velocity. A horizontal force P is pressing the block against the wall. What is the magnitude of the frictional force acting on the block?



(A) P (B) Mp

(C) mg (D) Zero

10. A piece of a highly reactive metal 'X' is dropped into a beaker of cold water. It reacts so violently that it catches fire and moves on the surface of the water. The resulting solution turns red litmus blue. Identify 'X'.

(A) Magnesium (B) Iron

(C) Sodium (D) Copper

11. During the processing of coal to obtain coke, a thick black liquid with an unpleasant smell is obtained. This liquid is a mixture of about 200 substances and is used as a starting material for manufacturing dyes and explosives. It is ___.

(A) Petroleum (B) Coal tar

(C) Coal gas (D) Bitumen

12. **Assertion:** Salt and edible oils are commonly used as preservatives in pickles.

Reason: These substances create an environment that promotes the growth of food-spoiling bacteria, making the pickle taste sour

(A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true, but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true, but Reason is false.

(D) Assertion is false, but Reason is true.

13. A tester is used to check the conduction of electricity through three different liquids: Vinegar, distilled water, and tap water. In which of these will the magnetic needle of the tester show a deflection?

(A) Vinegar and distilled water

(B) Tap water and distilled water

(C) Vinegar and tap water

(D) Only distilled water

14. Two musical instruments, a flute and a violin, play the same note at the same loudness. However, you can easily distinguish between the two sounds without seeing the instruments. Which characteristic of sound enables this distinction?

(A) Pitch (Frequency)
(B) Loudness (Amplitude)
(C) Timbre (Quality)
(D) Wavelength

15. A village forest has been cleared for a highway project. Within a year, the local farmers notice that the nearby river, which used to be clear, has become muddy and prone to frequent flash floods during rains. What is the primary ecological reason for this change?

(A) The construction of the highway added more soil to the river directly.

(B) The lack of tree cover increased the atmospheric temperature, causing more rain.

(C) Tree roots were holding the topsoil and assisting in the infiltration of water into the ground.

(D) Deforestation led to an increase in the population of herbivores that eroded the riverbanks.

16. In the human ear, which specific component acts as a stretched membrane that vibrates when sound waves strike it, transferring vibrations to the inner bones?

(A) Cochlea

(B) Eustachian tube

(C) Pinna

(D) Tympanic membrane (Eardrum)

17. Match the following alloys with their primary constituents:

Alloy	Primary Constituents
(i) Brass	(P) Iron, Nickel, Chromium
(ii) Bronze	(Q) Copper, Zinc
(iii) Stainless Steel	(R) Copper, Tin

(A) (i) → Q, (ii) → R, (iii) → P

(B) (i) → R, (ii) → Q, (iii) → P

(C) (i) → P, (ii → Q, (iii) → R

(D) (i) → Q, (ii) → P, (iii) → R

18. A boy runs towards a plane mirror with a speed of 2 m/s. At the same time, the mirror is moving away from the boy with a speed of 1 m/s. The boy is 10 m away from the mirror initially. After how much time will the boy meet his image?

(A) 2 s (B) 10 s

(C) 3 s (D) 5 s

19. A doctor finds that a patient has a high fever and a significantly low red blood cell count. Microscopic analysis of the blood shows a tiny, single-celled protozoan inside the red blood cells. Which of the following is the most likely way the patient contracted this disease?

(A) By consuming contaminated water containing Amoeba.

(B) By the bite of a female Anopheles mosquito.

(C) By inhaling droplets from an infected person coughing.

(D) By eating undercooked poultry infected with bacteria.

20. During a laboratory experiment, a student treats a plant cell with a chemical that specifically dissolves the "Middle Lamella." What is the most likely observation under the microscope?

(A) The cell membrane will burst, spilling the cytoplasm.

(B) Adjacent cells will start to separate from each other.

(C) The nucleus will disappear.

(D) The cell will lose its ability to perform photosynthesis.

21. Which of the following statements explains why a rainbow is always formed in the direction opposite to the Sun?

(A) The Sun's rays must hit the raindrops from behind the observer to undergo total internal reflection and dispersion.

(B) The raindrops act as mirrors and reflect the sunlight directly back to the sun.

(C) The Sun blocks the view of the rainbow if it is in front.

(D) Atmospheric refraction prevents the rainbow from forming near the Sun.

22. A student burns a small amount of sulphur powder in a deflagrating spoon and collects the gas produced in a gas jar. He adds a little water to the jar and shakes it. What will be the nature of the resulting solution when tested with litmus paper?

(A) It will turn red litmus blue because it is basic.

(B) It will turn blue litmus red because it is acidic.

(C) There will be no change because it is neutral.

(D) The litmus paper will be bleached white.

23. The process of slow conversion of dead vegetation into coal under high pressure and temperature over millions of years is called ____.

(A) Refining (B) Carbonization

(C) Vulcanization (D) Distillation

24. A pendulum oscillates 40 times in 4 seconds. Calculate its time period and frequency.

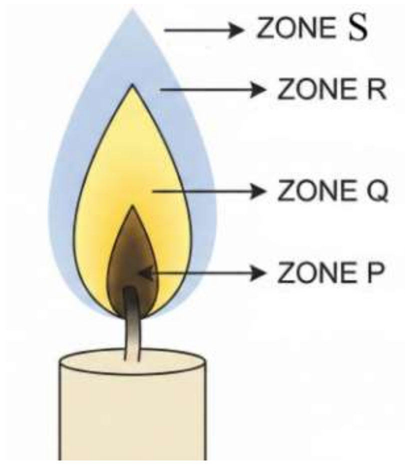
(A) Time Period = 0.1 s, Frequency = 10 Hz

(B) Time Period = 10 s, Frequency = 0.1 Hz

(C) Time Period = 0.01 s, Frequency = 100 Hz

(D) Time Period = 4 s, Frequency = 40 Hz

25. A student places a thin iron wire separately in Zone P, Zone Q, and Zone R of the flame for the same amount of time. In which zone the wire becomes red hot only?



(A) Zone R (B) Zone Q

(C) Zone P (D) Zone S

26. A boy standing near a cliff shouts and hears the echo 3 s later.
If the speed of sound in air is 330 m/s, the distance of the cliff from the boy is:

(A) 330 m (B) 495 m
(C) 660 m (D) 990 m

27. A scientist discovers a new unicellular organism in a volcanic hot spring. It has a cell wall and genetic material, but the genetic material is not enclosed in a nuclear membrane. Which of the following is true about this organism?

(A) It is a eukaryotic organism that has adapted to high temperatures.

(B) It is a multicellular organism whose cells have fused together.

(C) It is a prokaryotic organism, likely belonging to the group of bacteria.

(D) It is a virus that has developed a cell wall for protection.

28. In 1800, William Nicholson showed that passing an electric current through acidified water using electrodes produces gas bubbles. These bubbles were identified as oxygen and hydrogen. If the electrodes are connected to a 9V battery, on which electrode will the oxygen bubbles form?

(A) The electrode connected to the negative terminal.

(B) The electrode connected to the positive terminal.

(C) Oxygen bubbles will form on both electrodes equally.

(D) No bubbles will form if the water is pure.

29. Which of the following is stored under high pressure and is considered a cleaner fuel for transport vehicles?

(A) LPG (B) CNG

(C) Coal gas (D) Bitumen

30. A researcher is studying a specific cell and observes that it contains a rigid outer layer, a large central vacuole, and several green-colored organelles. However, when she treats the cell with a chemical that inhibits the function of the "Rough Endoplasmic Reticulum," she notices that the production of certain molecules stops immediately. Which molecules are most likely affected?

(A) Starch molecules used for energy storage in the vacuole.

(B) Lipid molecules used to build the cell membrane.

(C) Protein molecules that would have been exported or used in the cell.

(D) Cellulose molecules used to strengthen the cell wall.

31. A vaccine is introduced into a healthy person's body to provide immunity against a viral disease like Polio. What is the most accurate description of how this vaccine works inside the human body?

(A) The vaccine contains powerful antibiotics that wait in the blood to kill the virus if it enters.

(B) The vaccine provides a coating over all cells so the virus cannot attach to them.

(C) The vaccine introduces dead or weakened microbes that stimulate the body to produce antibodies, which remain in the body as "memory cells."

(D) The vaccine alters the person's DNA so that the virus no longer recognizes the human host.

32. A couple wants to have a child but faces issues with blocked Fallopian tubes. The doctor suggests a technique where the egg and sperm are fused in a laboratory and the zygote is then transferred to the mother's uterus. What is this baby commonly called?

(A) A cloned baby

(B) A test-tube baby

(C) A transgenic baby

(D) An asexual baby

33. Consider a heavy wooden box placed on the floor. Which of the following statements about the "Interaction" of forces is TRUE regarding this box?

(A) A force can only act on the box if it is moving.

(B) The box exerts no force on the ground because it is stationary.

(C) Force comes into play only when at least two objects interact; here, the box and the Earth (gravity) or the box and the floor (contact).

(D) If you push the box and it doesn't move, no force is being applied.

34. A student wanted to electroplate an iron key with copper to prevent it from rusting. He used a copper plate and an iron key as electrodes in a copper sulphate solution. Which of the following setups should he use?

(A) Connect the iron key to the positive terminal and the copper plate to the negative terminal.

(B) Connect both electrodes to the positive terminal of the battery.

(C) Connect the copper plate to the positive terminal and the iron key to the negative terminal.

(D) Use distilled water without any acid or salt as the solution.

35. If a huge area of forest is cleared for an industrial project, how does this specifically contribute to "Global Warming"?

(A) The machines used for clearing release a lot of heat into the atmosphere.

(B) The lack of trees means less Carbon Dioxide is absorbed from the atmosphere for photosynthesis, leading to an increased greenhouse effect.

(C) Deforestation increases the amount of Oxygen, which traps more heat from the sun.

(D) Fallen trees release a cold gas that reacts with the ozone layer

36. A security guard uses a large mirror to monitor a wide area of a shop. He notices that all the images in the mirror are erect (upright) but smaller than the actual objects, regardless of how far the objects are from the mirror. What type of mirror is this?

(A) Plane mirror

(B) Concave mirror

(C) Convex mirror

(D) Cylindrical mirror

37. Speed of light in a vacuum is 3×10^8m/s. If the refractive index of diamond is 2.42, what is the approximate speed of light inside a diamond?

(A) 1.24×10^8m/s

(B) 2.42×10^8m/s

(C) 0.8×10^8m/s

(D) 3.0×10^8m/s

38. Which property of metals is primarily utilized when manufacturing thin overhead cables for power transmission?

(A) Malleability

(B) Sonorous nature

(C) Ductility

(D) High melting point

39. **Assertion:** Biodiversity is essential for the survival of all life forms on Earth.
Reason: A variety of plants and animals ensures a stable food chain and maintains the balance of oxygen and carbon dioxide in the atmosphere.

(A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are true, but Reason is NOT the correct explanation of Assertion.

(C) Assertion is true, but Reason is false.

(D) Assertion is false, but Reason is true.

40. A 10 kg box is placed on a horizontal floor where the coefficient of static friction is 0.5. A boy pushes the box with a horizontal force of 30 N. What is the magnitude of the frictional force acting on the box? (Take g = 9.8 m/s^2)

(A) 49 N (B) 30 N

(C) 0 N (D) 98 N

41. A farmer noticed that his soil had become very hard and lacked aeration after several seasons of intensive farming. He decided to use a method that involves growing different crops in a pre-planned succession on the same field. He first grew Leguminous plants like peas, followed by Wheat in the next season. How does this specific practice primarily benefit the soil and the subsequent cereal crop?

(A) It increases the water-holding capacity of the soil by adding humus through the pea plants.

(B) Rhizobium bacteria in the root nodules of peas fix atmospheric nitrogen, replenishing soil nutrients for the wheat.

(C) Leguminous plants compete with weeds, ensuring that the wheat crop gets all the sunlight.

(D) The pea plants release specific acids that break down hard rocks into smaller soil particles faster than chemical fertilizers.

42. When a cracker is ignited, a sudden reaction takes place with the evolution of heat, light, and sound. This type of combustion is known as ____.

(A) Rapid combustion

(B) Spontaneous combustion

(C) Explosion

(D) Slow combustion

43. Which constituent of petroleum is commonly used for surfacing roads?

(A) Paraffin wax (B) Bitumen

(C) Lubricating oil (D) Diesel

44. A person accidentally steps on a rusty nail and is immediately given a "Tetanus" injection. This is different from a regular vaccine because it contains

"Readymade Antibodies." This type of protection is called:

(A) Active Immunity

(B) Passive Immunity

(C) Natural Immunity

(D) Permanent Immunity

45. A see-saw is balanced with a child of weight 300 N sitting 2 meters from the pivot on the left side. To balance the see-saw, where must a child of weight 600 N sit on the right side?

(A) 4 meters from the pivot.

(B) 1 meter from the pivot.

(C) 2 meters from the pivot.

(D) 0.5 meters from the pivot.

46. Two identical cotton cloth strips are taken. One cloth strip is dry, while the other is soaked in water and gently squeezed. Both cloth strips are brought one by one near a candle flame. It is observed that the dry cloth catches fire quickly, whereas the wet cloth does not burn. Which conclusion best explains this observation?

(A) Water reacts chemically with cotton and prevents burning.

(B) Wet cloth has no ignition temperature.

(C) Water absorbs heat and prevents the cloth from reaching its ignition temperature.

(D) Cotton cloth burns only in dry air.

47. If we compare a cell to a factory, which organelle would be the "Power House" and which would be the "Control Center"? Choose the correct pair.

(A) Mitochondria; Nucleus

(B) Nucleus; Chloroplast

(C) Ribosome; Mitochondria

(D) Cell Membrane; Vacuole

48. In an experiment, a scientist removed the pituitary gland of a young tadpole. He observed that the tadpole grew in size but never turned into a frog. What is the most likely reason for this?

(A) The pituitary gland produces the food required for metamorphosis.

(B) The pituitary gland stimulates the thyroid to produce thyroxine, which is essential for metamorphosis.

(C) The tadpole became too heavy to move onto land.

(D) Without the pituitary gland, the tadpole could not breathe air.

49. The most abundant metal found in the earth's crust is _____.

(A) Iron (B) Silicon

(C) Aluminium (D) Oxygen

50. A developer wants to build a factory on land that is currently "Fallow." What does "Fallow" land mean in agriculture?

(A) Land that is permanently infertile.

(B) Land left uncultivated for one or more seasons to regain its fertility naturally.

(C) Land that is used only for growing flowers.

(D) Land that has been flooded with water for rice cultivation.

KEY

1)	B	2)	C	3)	C	4)	B	5)	A	6)	C	7)	B
8)	B	9)	C	10)	C	11)	B	12)	C	13)	C	14)	C
15)	C	16)	D	17)	A	18)	B	19)	B	20)	B	21)	A
22)	B	23)	B	24)	A	25)	B	26)	B	27)	C	28)	B
29)	B	30)	C	31)	C	32)	B	33)	C	34)	C	35)	B
36)	C	37)	A	38)	C	39)	A	40)	B	41)	B	42)	C
43)	B	44)	B	45)	B	46)	C	47)	A	48)	B	49)	C
50)	B												